Beyond the prompt: Designing scalable AI systems

Google Developer Experts

About Me

- Leading AI Innovation **@66degrees**
- GDE and Top Voice AI and Cloud
- Top GenAI Community Contributor
- Content Creator, **40 M+ views**
- Speaker at IITs, IIMs, IIITs, MHA etc
- AI Advisor to GoI and Saudi Govt



The Architect's Mindset

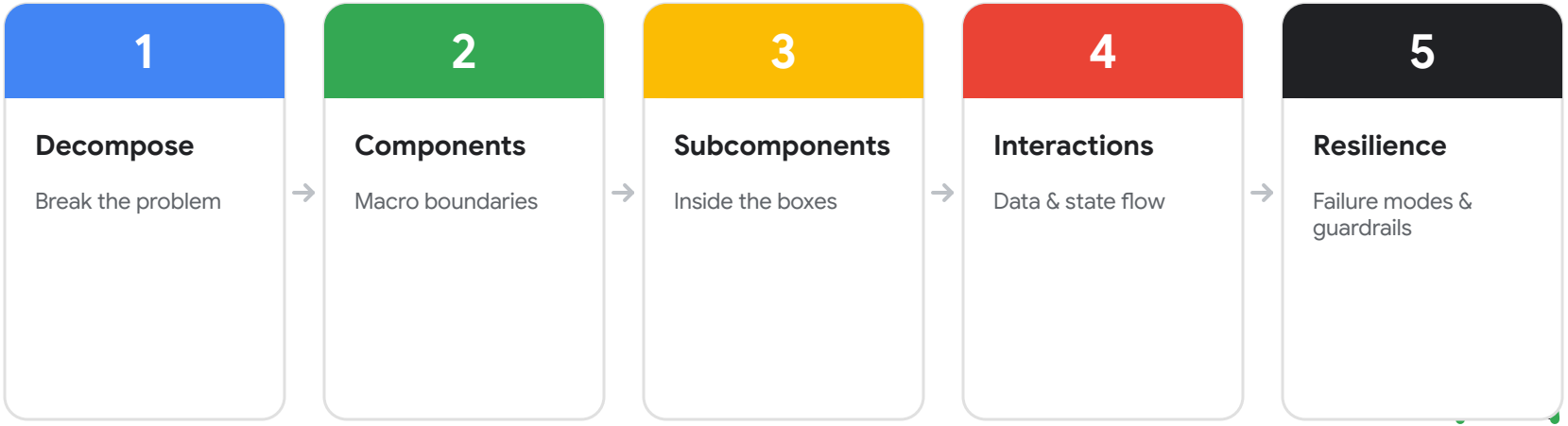
Prototype Thinking

Prompt -> LLM -> Output

System Architect Thinking

Deterministic Software + Probabilistic Agents

The 5-Layer System Design Framework



LIVE ARCHITECTURE WALKTHROUGH

Case Study: Real-Time UPI Fraud & Scam Resolution System



- **Latency:** <200ms Payment Check
- **Privacy:** Zero Raw PII / RBI Compliant
- **Scale Cost:** 500M+ Daily Txns

LIVE ARCHITECTURE WALKTHROUGH

Step 1: Decompose — Split Deterministic vs. Probabilistic

DETERMINISTIC SOFTWARE (<50ms, $0.0001 cost)

1. Check Velocity & Blacklist Rules → 2. Execute Safe Payment

AGENTIC AI (2-4 sec, Triggered only on anomalies/disputes)

3. Analyze Scam Context & Screenshot → 4. Negotiate Lien & Resolution

Rule #1 of Cloud Cost & Latency: 95% of traffic never touches an LLM.

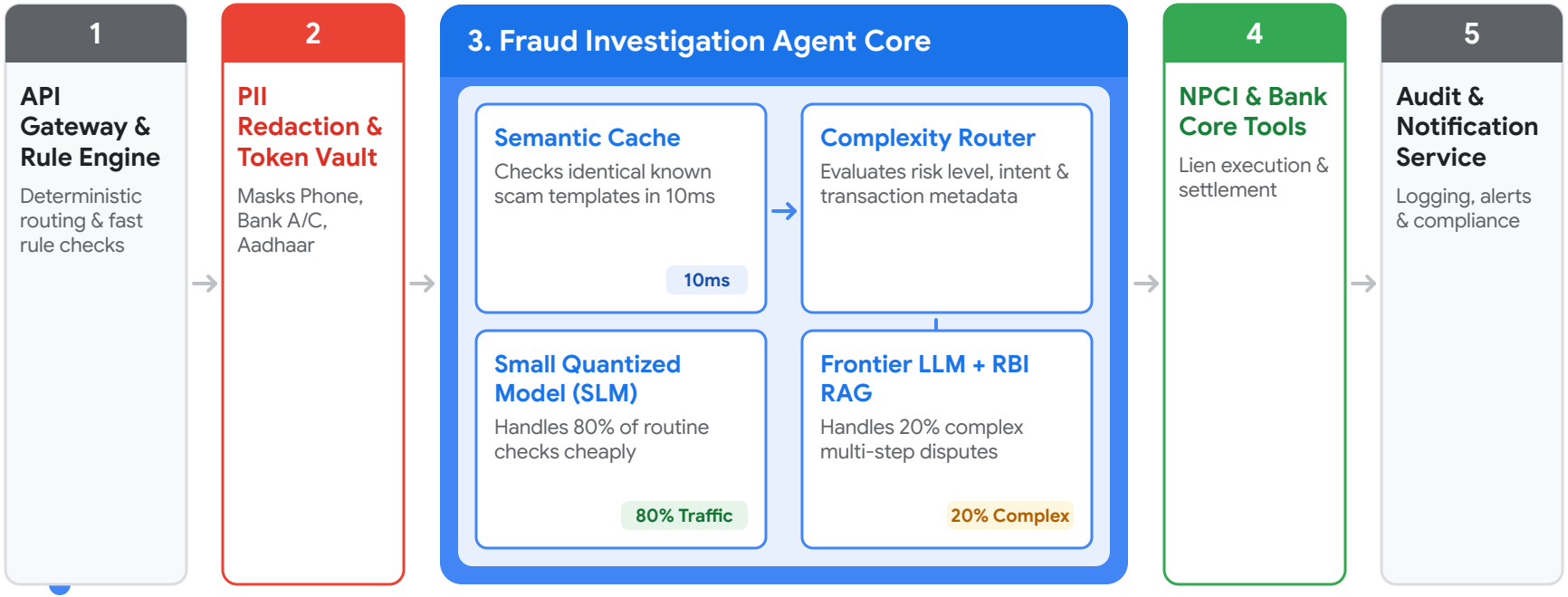
LIVE ARCHITECTURE WALKTHROUGH

Step 2: Components — Isolate Privacy & Macro Boundaries



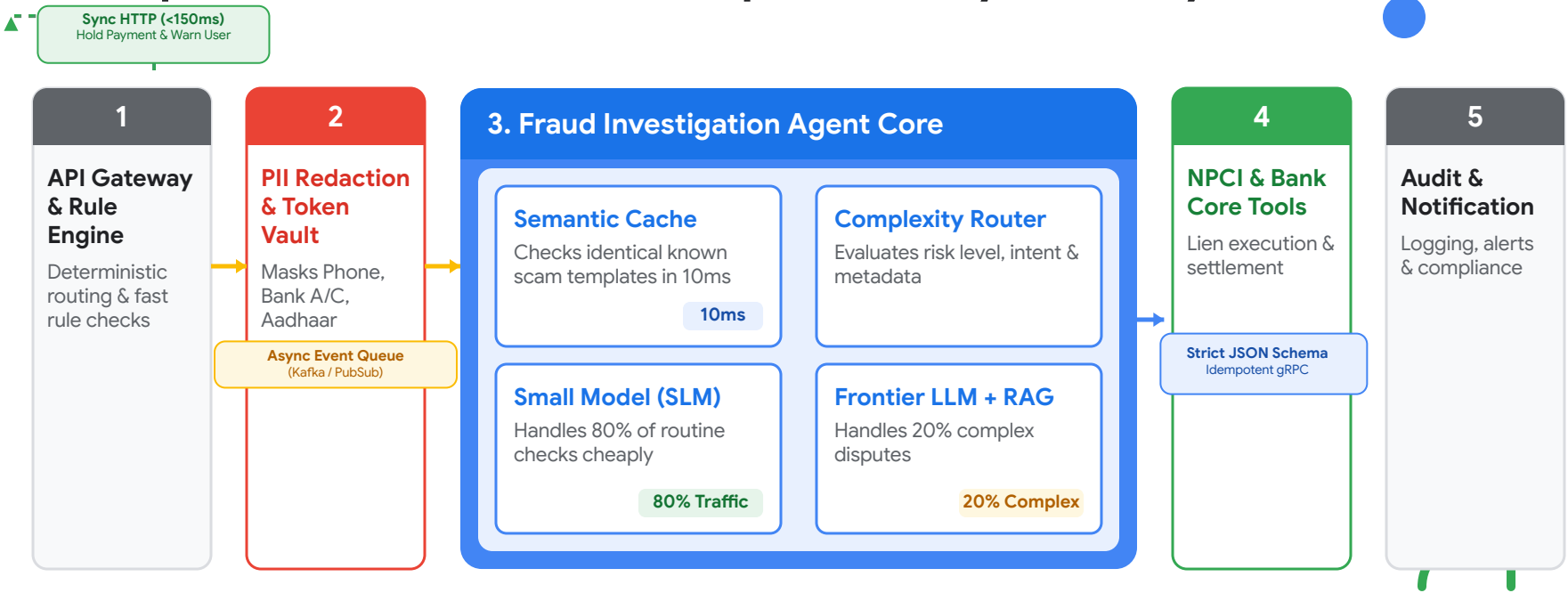
LIVE ARCHITECTURE WALKTHROUGH

Step 3: Subcomponents — Unpack the Agent for Cost & Speed



LIVE ARCHITECTURE WALKTHROUGH

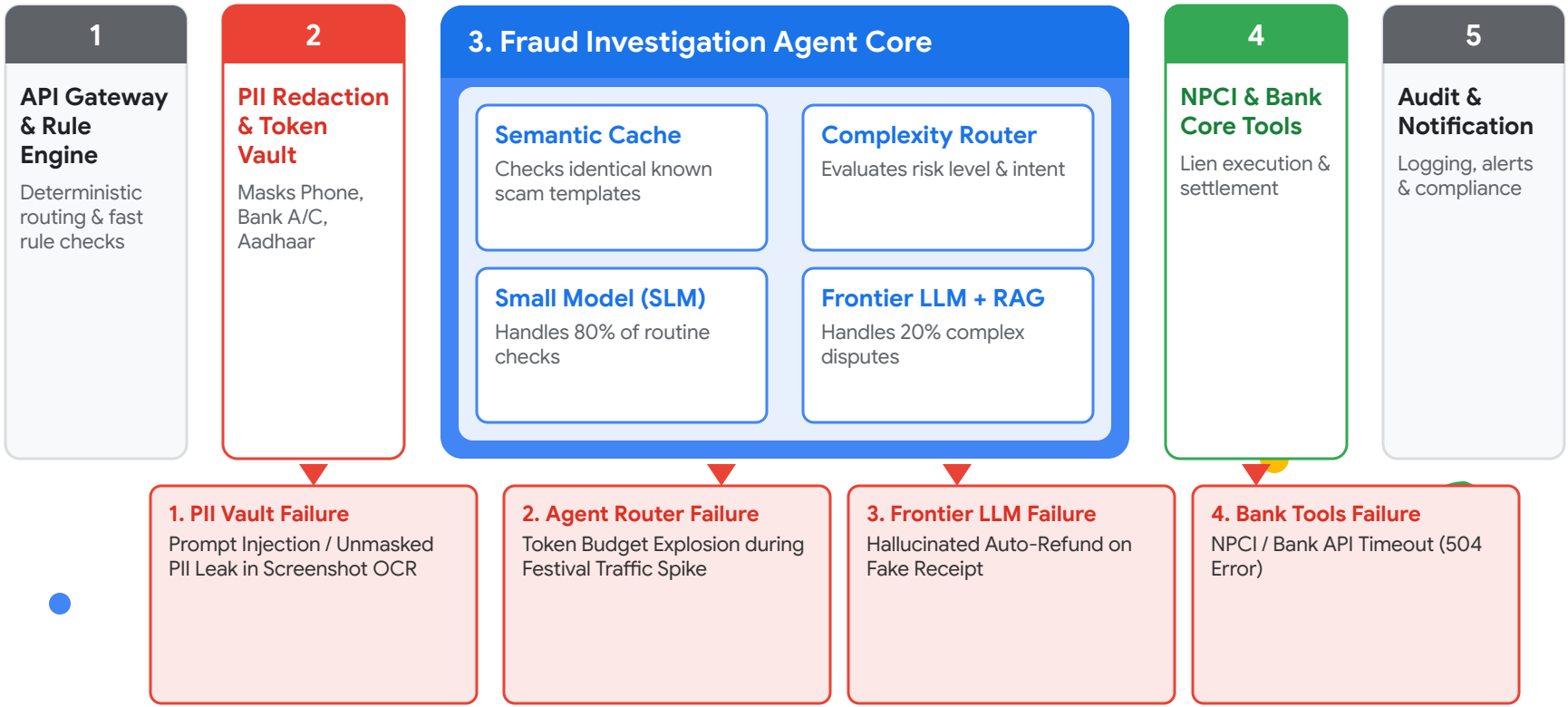
Step 4: Interactions — Decouple Latency with Async Flows



Key Principle: Never block a live user thread waiting for an Agentic reasoning loop.

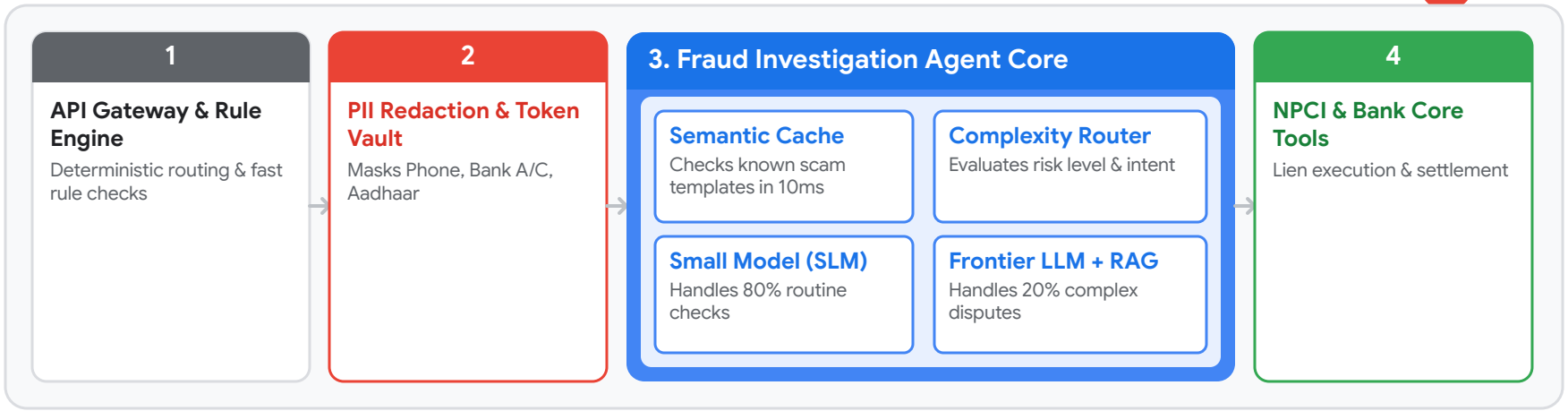
LIVE ARCHITECTURE WALKTHROUGH

Step 5A: Failure Analysis — Where Will Production Break?



LIVE ARCHITECTURE WALKTHROUGH

Step 5B: Resilience — The Final Hardened Architecture



1. Local On-Device OCR Masking

Redacts PII client-side before cloud upload, preventing leaks & saving bandwidth.

2. Token Budget Circuit Breaker

Caps Agent to max 3 reasoning loops per request to eliminate token budget explosions.

3. Deterministic Policy Guardrail

Hard safety rule sitting after Agent; strictly blocks auto-refunds exceeding ₹5,000.

4. HITL & Dead Letter Queue

Routes low confidence agent verdicts or bank API timeouts (504s) to human audit.

LIVE ARCHITECTURE WALKTHROUGH

How Our Architecture Evolved

V1: Naive AI

Single LLM Box
Direct monolithic LLM call for all incoming user requests

KEY METRICS & ISSUES

- 3.5s Avg Latency
- High Cloud Cost
- PII Exposed to Cloud

No safety or cost routing layer in place.

V2: Privacy-First

Deterministic Filter + PII Redaction Vault
Masks Phone, Bank A/C, Aadhaar before cloud

LLM Core Engine
Processes scrubbed input

KEY IMPROVEMENTS

- ✓ Zero PII Leaks
- 3.6s Latency
- High Cloud Cost

Added compliance guardrails at the expense of slight redaction overhead.

V3: Cost-Optimized

Semantic Cache (10ms)
Instantly hits known fraud patterns

Tiered Router (SLM / LLM)

- 80% traffic -> Small Model
- 20% traffic -> Frontier LLM

KEY IMPROVEMENTS

- ✓ 85% Cost Reduction
- ✓ 800ms Avg Latency
- Unhandled Edge Failures

Drastically cheaper & faster; still vulnerable to timeouts & loops.

V4: Production-Grade

Async Queues & Circuit Breakers
Decouples live user thread

Policy Guardrail (₹5k Cap)
Strict safety rules post-agent

HITL & Dead Letter Queue
Human fallback on 504/low confidence

PRODUCTION BENCHMARK

- ✓ <150ms Sync UX
- ✓ 99.9% Safe / Zero Hallucination
- ✓ Resilient to API Timeouts

Fully hardened enterprise architecture.

LIVE ARCHITECTURE WALKTHROUGH

The Architect's Checklist for Any Agentic System

- **Latency:** Put deterministic code on the critical path; push agents to async queues.
- **Privacy:** Strip and tokenize PII at the edge before any model sees the prompt.
- **Cost:** Filter with rules ➡ Hit semantic cache ➡ Route to Small Models ➡ Use Frontier LLMs last.
- **Reliability:** Wrap probabilistic agents inside deterministic guardrails and circuit breakers.

Let’s Connect!



